

MODELING ASPI BEHAVIOR USING EXCHANGE RATE INFORMATION : EVIDENCE FROM THE COLOMBO STOCK EXCHANGE

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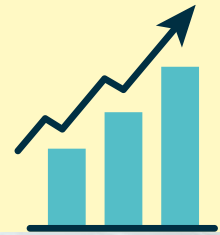


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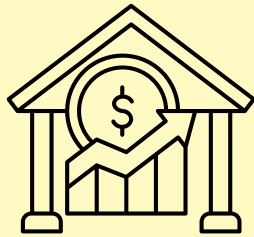
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Introduction



Stock Market



ASPI



Exchange Rate



Investors Decision



Problem Statement

- **Stock market prediction is complex**
- **Exchange rate influence unclear**
- **Need reliable model for ASPI**

prediction



Objectives



Main Objective

To analyze the relationship between ASPI and USD/LKR and develop a reliable model for predicting ASPI behavior



Specific Objectives

- **To analyze the relationship between ASPI and USD/LKR**
- **To develop predictive models for ASPI**
- **To compare the performance of different models**
- **To identify the most accurate forecasting model**



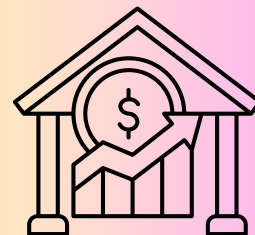
Data Description

Time Period



Sep 2024 – Sep 2025

ASPI



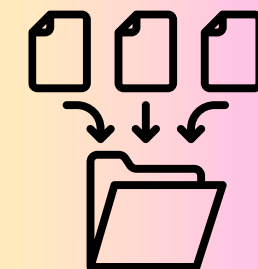
Colombo Stock Exchange

USD/LKR



Central Bank of Sri Lanka

Data



Daily time series data

Methodology

Data

**Descriptive
Analysis**

ADF Test

**Granger
Causality**

**Cointegration
& VECM**

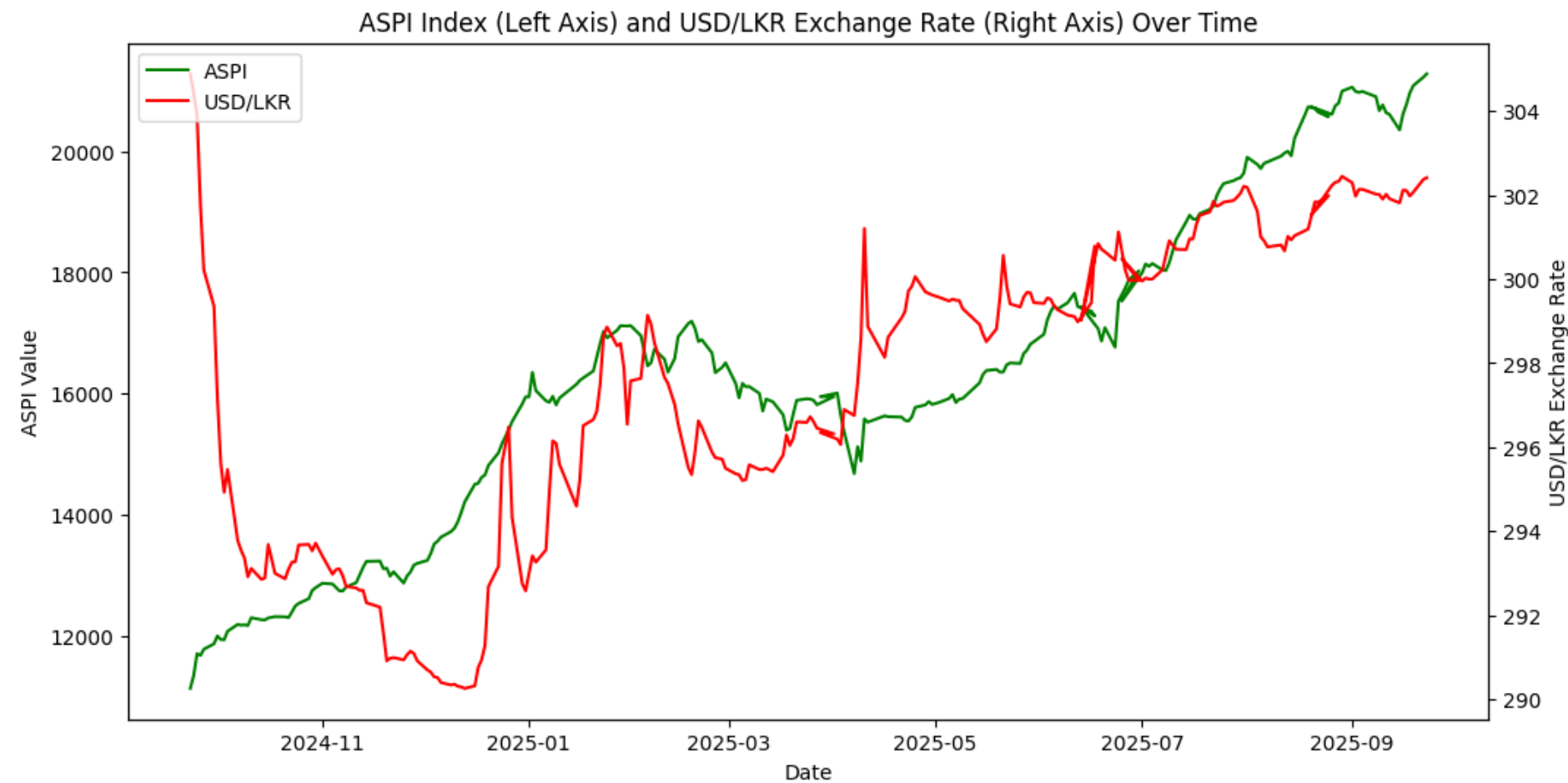
**Machine Learning
Models**

**Regression
Models**

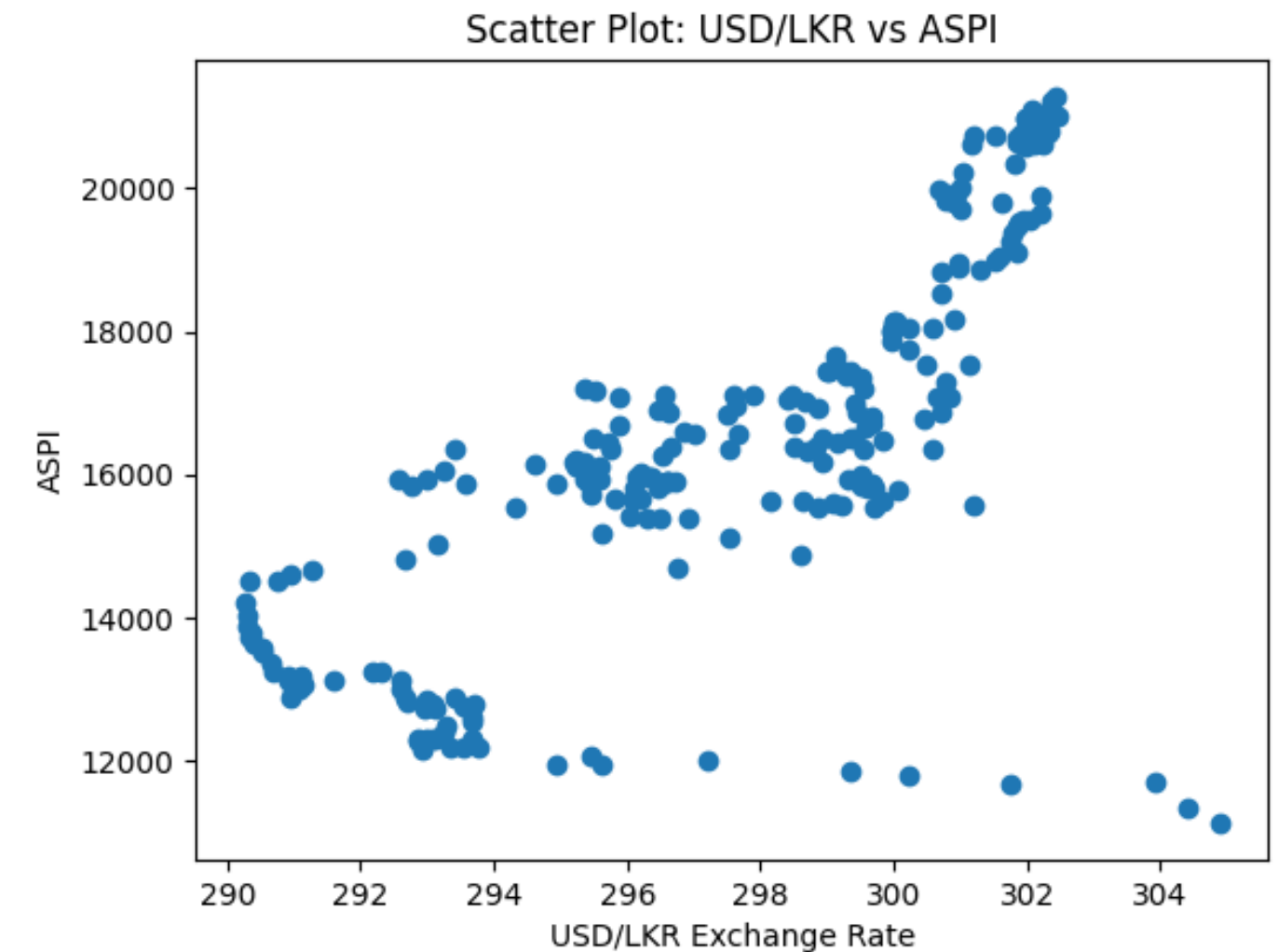


Exploratory Analysis

Trend of ASPI and USD/LKR Over Time



Relationship Between ASPI and USD/LKR



- **ASPI shows a clear upward trend with noticeable fluctuations over time**
- **USD/LKR exchange rate demonstrates a gradual increasing pattern**
- **Both variables exhibit similar movement patterns in the long run**

- **A strong positive relationship is observed between ASPI and USD/LKR**
- **The upward pattern indicates that higher exchange rates are associated with higher ASPI values**
- **Correlation coefficient confirms a strong positive relationship ($r = 0.73$)**

Stationarity and Granger Causality Results

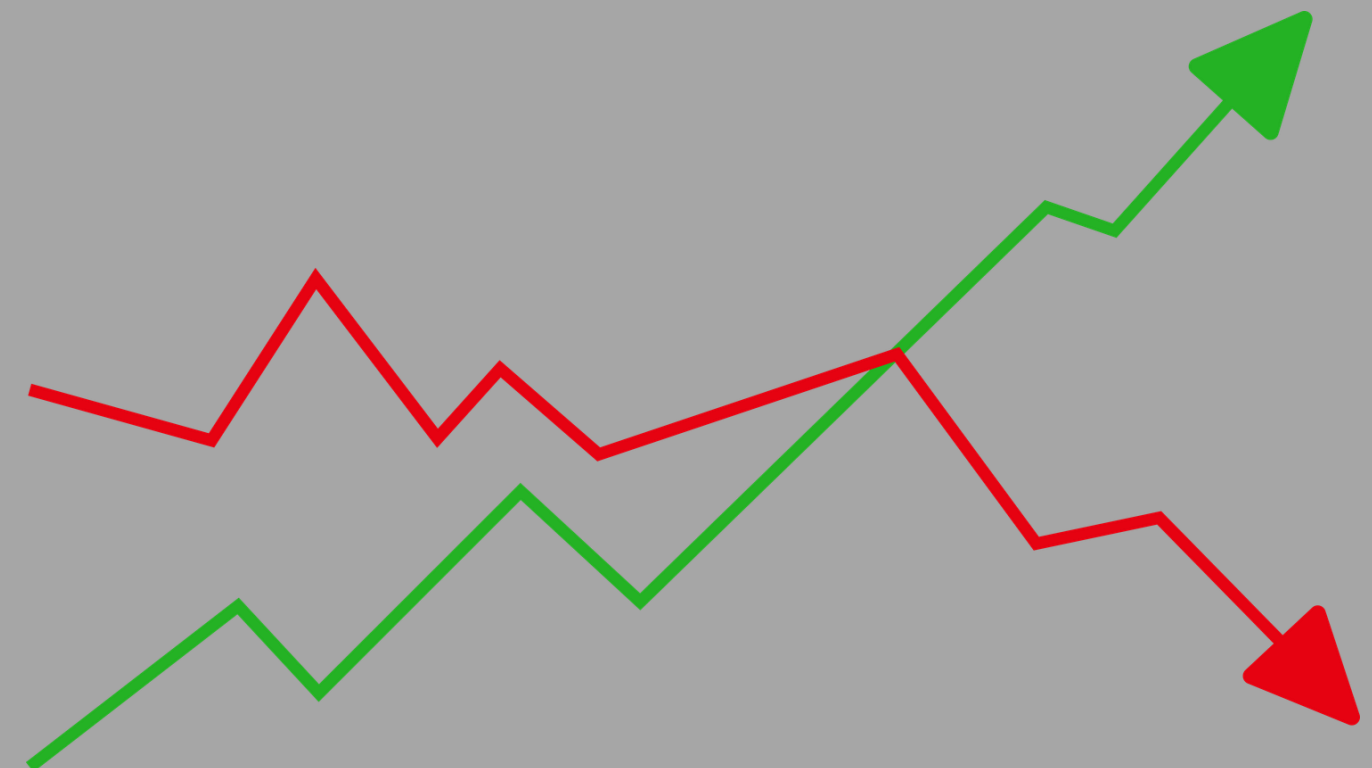
Stationarity Results

- ASPI and USD/LKR are non-stationary at level
- Both series become stationary after first differencing
- Therefore, both variables are integrated of order $I(1)$



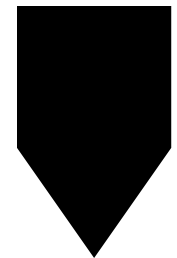
Granger Causality Results

- ASPI $\rightarrow$ USD/LKR ✓ Significant
- USD/LKR $\rightarrow$ ASPI ✗ Not significant
- Indicates a unidirectional relationship

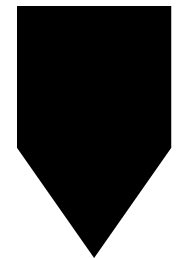


Cointegration and VECM Results

Cointegration Results

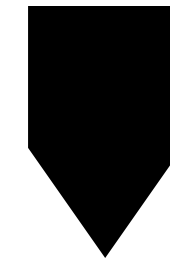


A long-run relationship exists between ASPI and USD/LKR

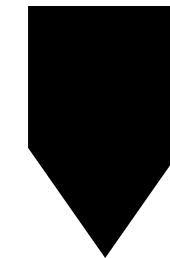


Variables move together over time

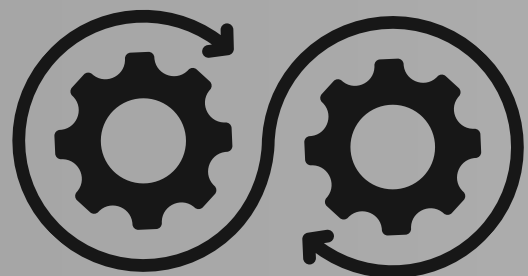
VECM Results



ASPI adjusts towards long-run equilibrium



Exchange rate shows weaker adjustment



Exchange rate has a limited but supportive role in explaining ASPI

Long-run relationship exists, but ASPI is primarily self-driven

Regression Models

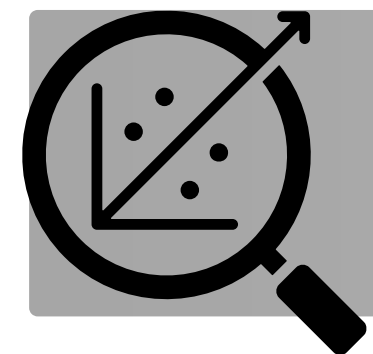
Model Approach

- Lag-based regression models were developed
- Used ASPI and USD/LKR lagged variables
- Estimated using Ordinary Least Squares (OLS)

Variables Used

- ASPI (dependent variable)
- ASPI_{t-1} (lagged ASPI)
- USD/LKR_{t-1} (lagged exchange rate)

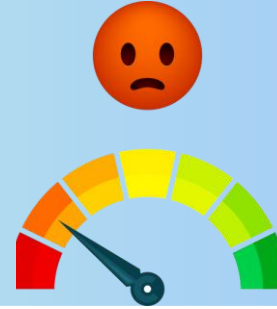
$$\text{ASPI}_t = \beta_0 + \beta_1(\text{ASPI}_{t-1}) + \beta_2(\text{USD}_{t-1})$$



ASPI lag is highly significant | Strong persistence in ASPI | Exchange rate has weaker effect

Machine Learning Models

Random
Forest



Poor

Ridge
Regression



Good

Lasso Regression



Best



Lasso gives highest accuracy | Handles multicollinearity | Best overall model

Lasso Forecast Example

Lasso Model:

$$\text{ASPI}_t = -1932.41 + 7.45(\text{ASPI}_{\text{Lag1}}) + 0.98(\text{USD}_{\text{Lag1}})$$

Example
:

Date	ASPI _{Lag1}	USD _{Lag1}	Predicted ASPI	Actual ASPI
2024.10.10	12177.35	293.36	12243.77	12165.88
2025.04.25	15613.1	299.82	15674.91	15770.64
2025.09.18	20775.4	302.10	20774.84	20965.30

Both variables influence ASPI prediction | Exchange rate shows a stronger effect in Lasso | Model simplifies prediction using key variables

Key Findings

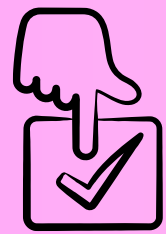


Key Findings

- Strong positive correlation between ASPI and USD/LKR
- ASPI Granger-causes exchange rate (unidirectional relationship)
- Long-run relationship exists between variables
- Lagged ASPI is the most consistent predictor
- Exchange rate shows a secondary influence

ASPI is primarily self-driven

Conclusion



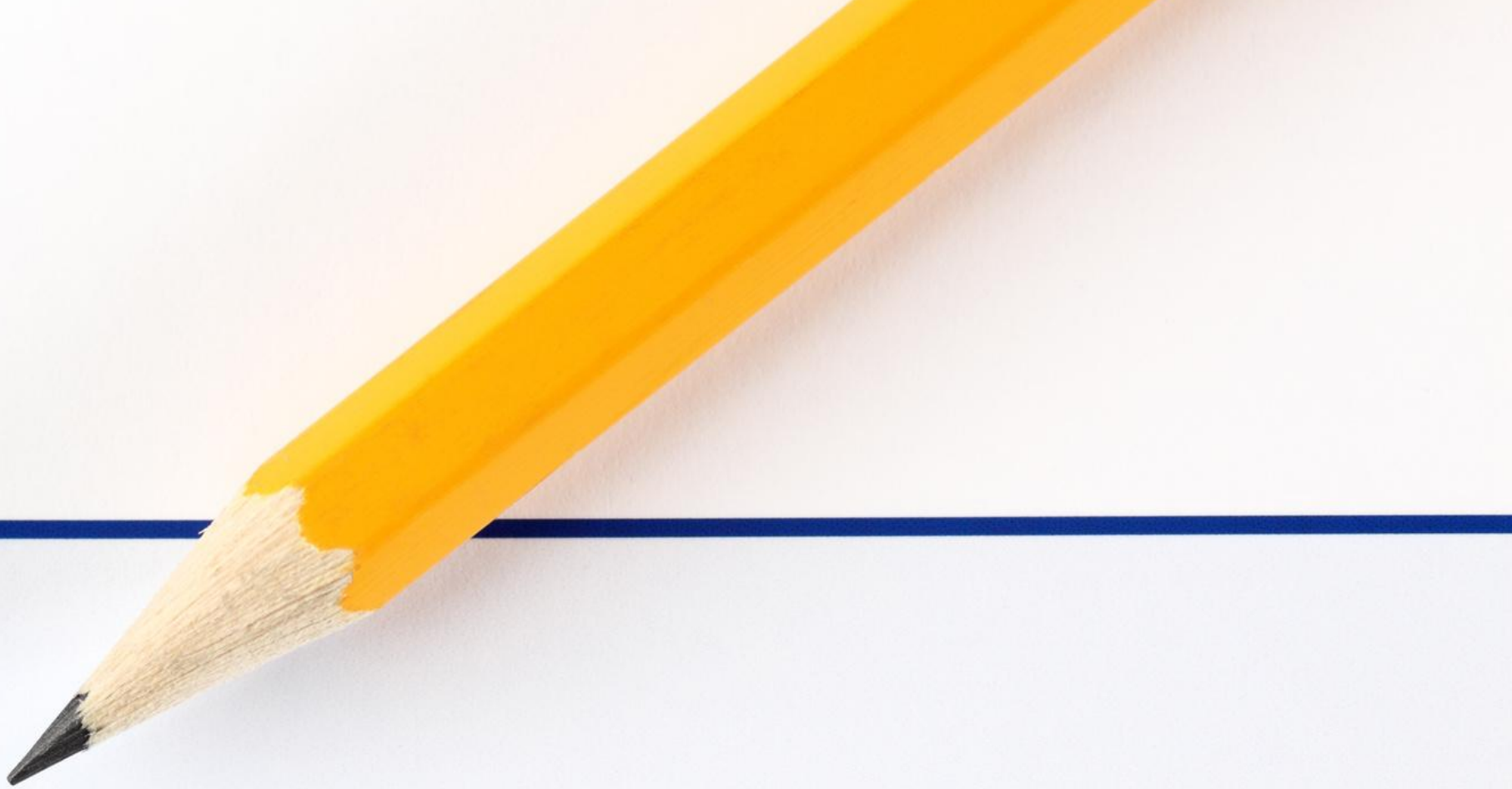
Conclusion

- ASPI behavior is mainly driven by its past values
- Exchange rate has a limited but supportive role
- Lag-based regression models provide stable and reliable results
- Lasso improves model simplicity through regularization

Lag-based models provide the most stable results

References

- Anusha, P., Dhushanthan, B., & Vinayagathan, T. (2022). The relationship between exchange rate and stock market performance: Empirical evidence from Sri Lanka. *Business and Economic Research*, 135–154.
- Balagobei, S. (2017). Macroeconomic variables and stock market returns in Sri Lanka. *Asian Journal of Finance and Accounting*, 118–132.
- Balagobei, S. (2022). Impact of macroeconomic variables on stock market performance: Evidence from Sri Lanka. *Wayamba Journal of Management*, 28–45.
- Balcilar, M. (2020). The foreign exchange and stock market nexus: New international evidence. *International Review of Economics & Finance*, 240–266.
- Gokmenoglu, K. E. (2021). Exchange rates and stock markets in emerging economies: New evidence using the quantile-on-quantile approach. *Quantitative Finance and Economics*, 94–110.
- Gooneratne, D. W. (2022). The effect of foreign exchange rates on stock market performance in Sri Lanka. *Journal of Management Matters*, 39–62.
- Jayasundara, J. M. (2019). Impact of macroeconomic variables on stock market performance: Evidence from Sri Lanka. *Sri Lankan Journal of Business Economics*, 1–18.
- Kengatharan, L. (2021). Macroeconomic variables and stock market performance in Sri Lanka: An ARDL bound testing approach. *South Asian Journal of Business Insights*, 43–58.
- Menike, L. M. (2010). The effect of macroeconomic variables on stock prices in the emerging Sri Lankan stock market. *Sabaragamuwa University Journal*, 50–67.
- Sujenthini, S. (2022). A study on effects of exchange rate fluctuation on stock returns: Evidence from Colombo Stock Exchange. *EPRA International Journal of Economics, Business and Management Studies*, 88–97.



THANK YOU